Data Science

Course 2026/27

1. Problem Description

Highway tolls were removed in Catalonia on September 1st 2021 for two major trunk highways (AP-7 and AP-2) and some secondary ones (a part of C-32 and C-33). There has been the feeling that making these highways free have changed the overall flow of cars in the Catalonia traffic network and the type of traffic accidents.

The goal of this project is to assess whether this intuition is true and, if that is the case, which are the major changes the highway toll removal brought into car accident typology.

In order to do so, you have access to all traffic accidents in Catalonia with the result of dead and/or severe injury in the period spanning from January 2010 to December 2024. The file and a description of its contents is available at this link.

In order to achieve the project goal each team should:

- Explore the data to understand whether there any traffic accident characteristic changes before and after tolls removal.
- Build a classifier that predicts whether an accident happened before or after September 1st 2025 and if possible, explains the drivers of such classification.

2. Tips

One of the issues of the problem at hand is that the class of accidents happening after September 1st 2021 are underrepresented compared to those that happened before. You need to look for strategies to alleviate this situation.

Changes in traffic accident type before and after September 2021 may be explained by the removal of tolls but also by the changes in traffic patterns and behavior as time goes by and a territory, its population distribution and habits changes.

On top of the available variables, teams may generate new features (a.k.a derived variables) that help improve the data analysis.

Teams will need to set up the validation strategy for the classifier they deem necessary and use the performance metrics that suits the problem best.

3. Schedule

The term project has to be delivered by **Friday, November 27th** at the task included in Atenea.

Teams will set up a meeting with the assigned professor to assess the progress on the project during May. In this meeting teams should come with a rough exploratory data analysis performed and bring any questions they may have.

4. Report Guidelines

Not including the title page, the table of contents and appendix, the report has to have at least 11 pages and no more than 14 (see breakdown below) single spaced and font size 11.

The R code created to perform the analyses must be included in an appendix or zip file (it does not count in the total length of the report).

When writing the report, think about what you want to convey to the reader. Information, graphs and results should be used to validate your message. Do not include information for the sake of it!

6. Term Project. Outline.

I. TOC

Include a Table of Contents.

II. Executive summary (max. 1 page)

One page summary of the project. What anybody without much time needs to read to get an idea of the project goals, methodology and results. This is a very important part in any data science project. It is what 95% of the people will ever read of your project.

III. Project Structure (max. 1 page)

Describe the project goal and the approach you have used to reach it. Describe the methodology used for model building and validation.

IV. Data and exploratory analysis (3 to 4 pages)

Explain the data you used for the project. This includes listing the relevant response and explanatory variables (what they measure, units of measure, etc.) and the exploratory data analysis.

V. Model (3 to 4 pages)

Explain the model/s used to predict/understand the response variable. Indicate the tunning parameters from R used in the model building. Describe the model validation strategy.

VI. Results (2 to 3 pages)

Describe and justify the major findings of your project. What are its results? What conclusions can you draw from it? Make sure you answer the questions presented in the project goal. Justify your conclusions with graphs or outputs (parameter values, etc.) from the model build.

VII. Next steps (1 page)

Describe what would be your follow up on the project results. What things would you do from here on? What things would you do differently if you had to start again?

VIII. Annex (not included in the page count and given in a different file)

R Code